

LP3 (ISR) – Assignment 4 – Precision and Recall (RollNo: 43221)

Input Code:

```
import java.util.*;

public class PrecisionRecallCalculator {
    public static void main(String[] args) {
        // Sample input: Answer set A, Query q1, and Relevant documents Rq1
        Set<String> answerSetA = new HashSet<>();
        answerSetA.add("Doc1");
        answerSetA.add("Doc2");
        answerSetA.add("Doc3");
        answerSetA.add("Doc4");
        answerSetA.add("Doc5");

        Set<String> relevantDocumentsRq1 = new HashSet<>();
        relevantDocumentsRq1.add("Doc1");
        relevantDocumentsRq1.add("Doc2");
        relevantDocumentsRq1.add("Doc3");

        // Query q1
        String query = "q1";

        // Calculate precision and recall
        double precision = calculatePrecision(answerSetA,
relevantDocumentsRq1);
        double recall = calculateRecall(answerSetA, relevantDocumentsRq1);

        // Print the results
        System.out.println("Query: " + query);
        System.out.println("Precision: " + precision);
        System.out.println("Recall: " + recall);
    }

    // Calculate precision
    public static double calculatePrecision(Set<String> retrievedDocuments,
Set<String> relevantDocuments) {
        int relevantRetrieved = 0;

        for (String doc : retrievedDocuments) {
            if (relevantDocuments.contains(doc)) {
                relevantRetrieved++;
            }
        }

        return (double) relevantRetrieved / retrievedDocuments.size();
    }

    // Calculate recall
    public static double calculateRecall(Set<String> retrievedDocuments,
Set<String> relevantDocuments) {
        int relevantRetrieved = 0;

        for (String doc : retrievedDocuments) {
            if (relevantDocuments.contains(doc)) {
                relevantRetrieved++;
            }
        }
    }
}
```

```
        return (double) relevantRetrieved / relevantDocuments.size();  
    }  
}
```

Output:

"C:\Program Files\Java\jdk-22\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.1.4\lib\idea_rt.jar=58249:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.1.4\bin" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8 -classpath
F:\BE\LP3\GroupB_Assignment_1_PrecisionRecallCalculator\out\production\GroupB_Assignment_1_PrecisionRecallCalculator PrecisionRecallCalculator

Query: q1

Precision: 0.6

Recall: 1.0

Process finished with exit code 0